Power

VSYS comes from the 12 V input: buck to +5V, then D4 (Power sheet). D2 keeps that 5 V off the USB port; USB alone runs it through D2 (no +5V on ribbons).

CC1/CC2 5.1k pull-downs: a USB-C host sees a USB 2.0 device and turns on 5 V.

Radio (RM2)

RM2 gSPI: GPIO43 = WL_ON + BT_ON, GPIO44 = CS, GPIO45 = DATA (DI direct, DO through R14, IRQ through R15), GPIO46 = CLK. Same circuit as the Pico 2 W, moved off GPIO23/24/25/29; set CYW43_DEFAULT_PIN_* in the board header. Hand-solder U5 (not stocked at LCSC). Keep copper out from under the antenna end.

Make sure R7 and R8 are close to RP2350

12 V in, 5 V buck

Power

File: Power.kicad_sch

Primary Flash

If using RP2354 (internal flash), remove R10. R9 can be fit to use U3 as secondary memory.

BOOTSEL Button

RTC and headers

RTC: the DS3231 circuit from universal-strip-controller (U9), on I2C0: SDA = GPIO32, SCL = GPIO33, INT/SQW = GPIO34 (open drain, pulled up). 32KHZ and RST unused. CR1220 in a Keystone 3001 holder.

J3 SWD

J2 SPARE

PSRAM

8 MB PSRAM (U4) on the second QSPI chip select (XIP_SS_N[1]), moved from GPIO0 to GPIO47. R13 pull-up keeps it deselected at power-up (GPIOs default to pull-downs). Fit R11 = 0R.

12MHz Crystal

RUN Button

GPIO0-31 to the four output shrouds

Outputs

File: Outputs.kicad_sch

J2: spare GPIO35-42. UART1 TX/RX = GPIO36/37. GPIO40-42 = ADC0-2. J3: SWD in a Pico's debug pin order. GPIO0-31 go to the four output shrouds (Outputs sheet). GPIO43-47 are the RM2 and the PSRAM.

MH1 NPTH FD2.75mm SM3.15mm KO6mm
MH2 NPTH FD2.75mm SM3.15mm KO6mm
MH3 NPTH FD2.75mm SM3.15mm KO6mm
MH4 NPTH FD2.75mm SM3.15mm KO6mm

Note: S1 schematic revision indicates that this is the 1st revision of the schematic/BOM associated with the R4 PCB. R4-S1 therefore uniquely identifies the fully built PCB Assembly (PCBA). This is a change from the previous revision numbering to align with the internal Raspberry Pi product revisioning scheme

VARIANT	VARIANT1
REF:	RPI-RP2350B-MINIMAL
DESCR:	RP2350B QFN-80 Minimal Design Example

SCHEMATIC/BOM REVISION: S1

BOM is autogenerated from schematic, therefore schematic rev. == BOM rev. Keep Titleblock rev. in sync with above. Reset to 1 when creating new R num.

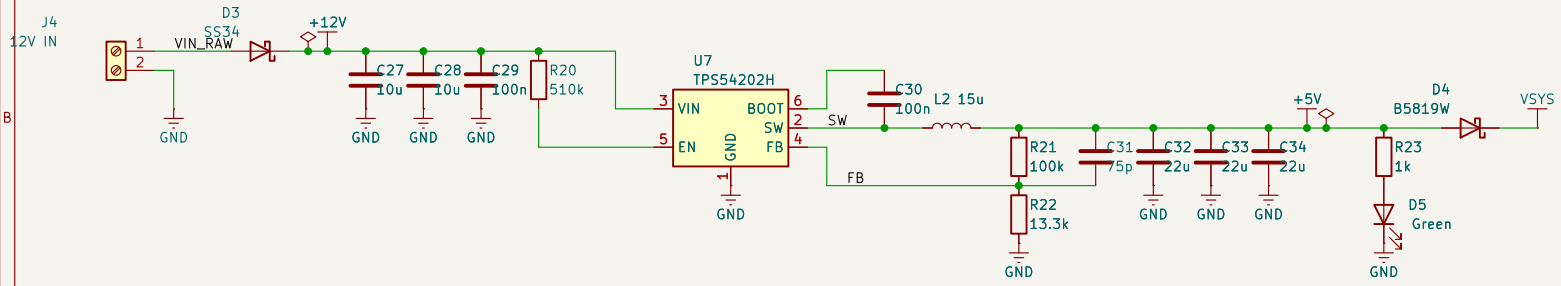
PCB/PWB REVISION: R4 <- PCB/PWB rev this schematic is for

MANUFACTURER: XXX
MANUF. PN: XXX
CIRCUITS/PANEL: XXX
DESCRIPTION: XXX

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TITLE	LED processor board (RP2350B + RM2)		
DATE	2026-10-14	REF	RPI-RP2350B-RM2 / RPI-RP2350B-MINIMAL
SIZE	A3	DRAWN	Original design: Mike Stimson, Raspberry Pi
REV	R4-S1	SHEET	1/3

12 V in, 5 V buck



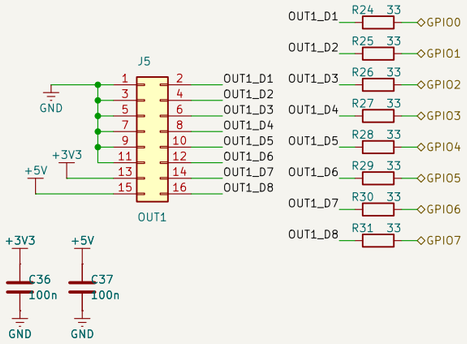
U7 values follow TI's TPS54202HEVM-716 5 V design (100k / 13.3k with 75 pF across the top resistor: 0.596 V x (1 + 100/13.3) = 5.08 V; 15 uH; 500 kHz). R20 enables U7 from the 12 V side. D3 blocks a reversed 12 V lead. The load is small: the 3.3 V regulator on VSYS (RP2350, RM2, flash, PSRAM) and the four output boards' shifters (VCCB on +5V, VCCA on +3V3), well under 0.5 A. +5V feeds the ribbons directly; VSYS gets it through D4 so USB can't back-feed the buck.

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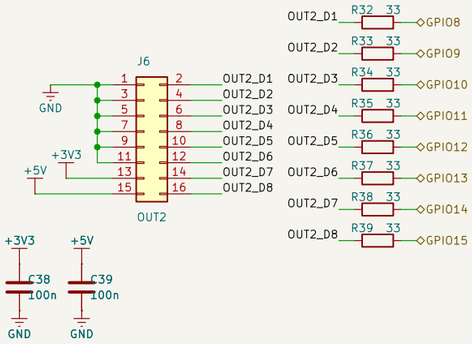
TITLE	LED processor board: Power				
DATE	2026-10-19	REF		REV	A
SIZE	A4	DRAWN		SHEET	2/3

Output shrouds: 16-pin IDC to the output boards. Pin 1 is the triangle on the shroud.
Odd pins 1-11 GND, 13 = +3V3 (shifter VCCA), 15 = +5V (shifter VCCB); data 1-8 on even pins 2-16
(same as led-output-board J2 and MegaTree docs/power.md). 33R series resistors at the RP2350 end.

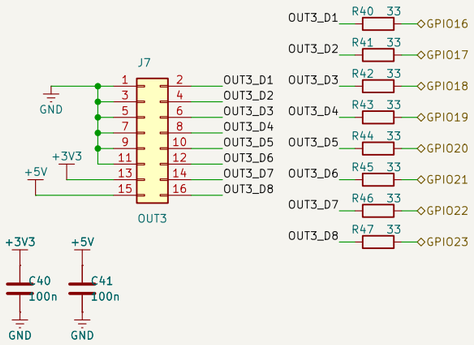
OUT1: GPIO0-7



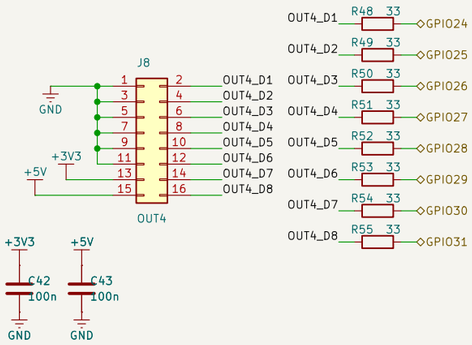
OUT2: GPIO8-15



OUT3: GPIO16-23



OUT4: GPIO24-31



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TITLE	LED processor board: Outputs				
DATE	2026-10-19	REF		REV	A
SIZE	A3	DRAWN		SHEET	3/3